

# A Dual Operator for Prime–Zero Coupling

## and a Conditional Proof of Energy Asymmetry

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### Abstract

We introduce the operator  $\tilde{T} := \Phi \circ \Phi^*$ , the dual of the loop operator  $T = \Phi^* \circ \Phi$  constructed in Paper 3 of this series. Here  $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$  maps a finite-dimensional prime space into a finite-dimensional zero space via Gaussian-damped prime resonance vectors.

**What is proved in this paper.** We establish (§2.3) the algebraic spectral identity  $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$  by classical operator theory; the maximum discrepancy between shared eigenvalues is below  $10^{-15}$  (machine precision). We prove (§3.4) that  $W_1 := C_T \cdot \tilde{T}^+$  is self-adjoint.  $W_1$  does not assume RH or the HP postulate; however, its normalization constant  $C_T$  is calibrated against the zero ordinates via OLS and should not be presented as fully input-free.  $\tilde{T}$  itself — built from prime resonance vectors without  $\gamma_k$  as algebraic inputs to the operator — is the circularity-free object (§3.4). We prove (§4.3, Lemma 4.3) the **Abel Summation Principle** (Lemma M3): for monotone decreasing weights  $f_i$  and a sequence  $g_i$  with bounded partial sums,  $|\sum_i f_i g_i| \leq M f_1$ . We prove (§4.4, Theorem 4.4) the **Prime Exponential Sum Bound** (Theorem EXPLICIT): for fixed  $\gamma > 0$ ,

$$M_k(\kappa) := \max_{x \leq \kappa} \left| \sum_{p \leq x} \sin(\gamma_k \log p) \right| = O\left(\frac{\pi(\kappa)}{\gamma_k}\right)$$

using only the Prime Number Theorem (no Riemann Hypothesis). We prove (§2.4, Proposition 2.11) that  $\tilde{T}$  is **not** a Hilbert–Pólya operator: the correlation  $r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$  falls from 0.50 to 0.16 for  $\kappa \in \{23, \dots, 503\}$ .

**What is numerical.** The eigenvectors of  $\tilde{T}$  localize at zero ordinates  $\gamma_k$  with degrees 0.45–0.99 [Numerical, §3.2]. The correlation  $r_1 = \text{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950$  ( $\kappa = 53$ ) [Numerical, §3.1]. The energy asymmetry  $\eta_{\text{orig}} > 0$  holds for all tested  $\kappa \leq 1009$  [Numerical, §5.1].

**What is conditional.** Under Assumption A — a condition on the line  $\text{Re}(s) = 1$ , distinct from the Riemann Hypothesis: (a)  $\zeta(1 + in\gamma_k) \neq 0$  for all  $n \geq 1, k \leq N$  (this follows directly from Hadamard–de la Vallée Poussin (1896), which establishes  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$ ; hence not genuinely open), and (b) joint equidistribution of prime log-phases (§6, Open Problem 6.1) (the genuinely open condition) — we prove (§5.3, Theorem 5.3(i)(ii)) that the normalised cross/stream averages vanish [CONDITIONAL on Assumption A]. We establish numerically (Theorem 5.3(iii)(iv)) that  $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$  for all tested  $\kappa > 7$  [NUMERICAL lower bound,  $\kappa$ -invariant] and  $\eta_{\text{orig}}(\kappa) \rightarrow \eta_{\infty} \approx 0.81 > 0$  [NUMERICAL].

**What remains open.** The analytic proof of  $\eta_{\text{orig}} > 0$  without the above assumption (Open Problem 6.2); whether a circularity-free spectral function  $f(\tilde{T})$  exists with spectrum converging to  $\{\gamma_k\}$  (Open Problem 6.3); and the arithmetic origin of  $C_{\eta} \approx 0.39$  (Open Problem 6.4).

**Structure.** The formal core is self-contained in §§ 2–5. Sections 6 and 7 contain open problems and a geometric interpretation (clearly marked, not used in proofs). No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–5.

# 1 Introduction

## 1.1 Background and Motivation

This paper is the fourth in a series studying the curvature of the Riemann zeta function from a geometric and operator-theoretic perspective. A reader familiar with Paper 3 [10] may begin at §2; the present introduction provides series context and an explicit reader contract for all results. The formal content in §§ 2–5 is self-contained.

**Paper 1** [8] introduced the local curvature function

$$H_{\xi}(\sigma, \kappa) := \sum_{p \leq \kappa} V_p(\sigma),$$

where  $V_p(\sigma) = \frac{d^2}{d\sigma^2} \log \|M_{\sigma} f_{\kappa,p}\|^2 > 0$ . A key finding was the divergence  $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2 \rightarrow \infty$  at  $\sigma = \frac{1}{2}$ , contrasted with  $H_{\text{local}}(\sigma, \kappa) \rightarrow C(\sigma) < \infty$  for all  $\sigma > \frac{1}{2}$ .

**Paper 2** [9] constructed the Weil functional and established  $W(g^* * \check{g}^*) = Z(g^*) - H_{\text{local}}(\sigma, \kappa) + O(\varepsilon)$ , connecting local curvature to the Weil positivity criterion.

**Paper 3** [10] made the geometric structure explicit. Given a prime cutoff  $\kappa$  and  $N$  zero ordinates  $\gamma_k$ , two finite-dimensional real Hilbert spaces  $H_{\text{str}}$  (over primes) and  $H_{\text{null}}$  (over zeros) are connected by a linear map  $\Phi$ . The loop operator  $T = \Phi^* \circ \Phi$  is self-adjoint and positive semi-definite. The energy asymmetry  $\eta_{\text{orig}} = \Delta/E_{\text{str}} \in [0.598, 0.704]$  (benchmark  $\kappa = 53$ ,  $\varepsilon = 0.05$ ,  $N = 100$ ) was established numerically; the structural energy  $E_{\text{str}} \rightarrow 6.1$  as  $\kappa \rightarrow \infty$  (analytically explained by absolute convergence of  $\sum_p (\log p)^2/p^2$ ).

**The present paper** introduces and analyzes  $\tilde{T} = \Phi \circ \Phi^*$ , the dual of  $T$ , acting on the zero space  $H_{\text{null}}$ . The operator emerges canonically from the same  $\Phi$  already constructed in Paper 3, and requires no new postulate.

The main conditional result of Paper 4 is a **conditional analytic proof** that  $\eta_{\text{orig}} > 0$ : under Assumption A — two conditions on  $\text{Re}(s) = 1$  — the energy asymmetry persists in the limit  $\kappa \rightarrow \infty$ . Both conditions concern  $\text{Re}(s) = 1$ , distinct from RH which concerns  $\text{Re}(s) = \frac{1}{2}$ . Non-vanishing on  $\text{Re}(s) = 1$  is established for all real  $t \neq 0$  by Hadamard and de la Vallée Poussin (1896); this covers the values  $t = n\gamma_k$  directly. The genuinely open condition is the equidistribution of  $\{\gamma_k \log p \bmod 2\pi\}$ , which is the main open problem of this paper.

## 1.2 Non-Circularity Statement

*The following holds unconditionally throughout §§ 2–5:*

### No result in this paper assumes:

- (i) The Riemann Hypothesis;
- (ii) The GUE conjecture;
- (iii) The Montgomery pair correlation conjecture;
- (iv) The Hilbert–Pólya postulate.

The operator  $\tilde{T} = \Phi \circ \Phi^*$  is constructed explicitly from the prime resonance vectors  $\{a_p\}_{p \leq \kappa}$  (defined in § 2.1). The zero ordinates  $\gamma_k$  appear as *inputs* to the construction of  $\Phi$ , not as conclusions. In Theorem 5.3 (§ 5.3), Assumption A is explicitly labelled **CONDITIONAL**; it is a condition on  $\text{Re}(s) = 1$ , distinct from RH which concerns  $\text{Re}(s) = \frac{1}{2}$ . Its logical relationship to RH is not analysed in this paper.

In particular,  $\tilde{T} = \Phi \circ \Phi^*$  is a deterministic, explicitly computable finite-dimensional operator — not a random matrix or a statistical ensemble. No GUE conjecture, no Montgomery pair correlation conjecture, and no random-matrix-theoretic assumption is used or implied anywhere in §§ 2–5. The structural similarity between eigenvector localization in  $\tilde{T}$  and GUE level statistics is a numerical observation, not a postulate.

*Note on  $W_1$ :* The inverse operator  $W_1 = C_T \cdot \tilde{T}^+$  does not assume RH or the HP postulate; however, its normalization constant  $C_T$  is calibrated against the zero ordinates via OLS (§ 3.3).  $W_1$  is therefore not fully input-free. Only  $\tilde{T}$  itself — built from prime resonance vectors without  $\gamma_k$  as algebraic inputs to the operator — is the circularity-free object of this paper.

### 1.3 Imported Results from Papers 1–3

The following results from earlier papers are used in this paper, exclusively in the indicated roles.

| Label | Result                                                              | Status                     | Used where                   |
|-------|---------------------------------------------------------------------|----------------------------|------------------------------|
| (R1)  | $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2$       | Proved [Paper 1, § 2]      | § 1.1, § 7 (motivation only) |
| (R2)  | Weil bridge                                                         | Proved [Paper 2, § 3]      | § 1.1, § 7 (motivation only) |
| (R3)  | $T = \Phi^* \Phi$ , $\eta_{\text{orig}} \in [0.598, 0.704]$ , $c_p$ | Proved/Numerical [Paper 3] | § 2.1, § 5                   |
| (R4)  | $E_{\text{str}}(\infty) \approx 6.1$ (convergent)                   | Numerical [Paper 3, § 5]   | § 5.1 (context)              |

Results (R1) and (R2) are used for motivation only. Results (R3) and (R4) are recorded for completeness; the formal core in §§ 2–5 depends only on the definitions in § 2.1, not on  $\eta$ -values from Paper 3.

## 1.4 What This Paper Proves and What It Imports

*Reader Contract.*

| Content                                                                                  | Status                         | Where |
|------------------------------------------------------------------------------------------|--------------------------------|-------|
| $\tilde{T} = \Phi \circ \Phi^*$ definition                                               | Proved here                    | § 2.2 |
| Self-adjointness and PSD of $\tilde{T}$                                                  | Proved here                    | § 2.2 |
| Algebraic spectral identity $\lambda_j(T) = \mu_j(\tilde{T})$                            | Proved here                    | § 2.3 |
| $\tilde{T}$ is NOT a Hilbert–Pólya operator                                              | Num. negative result           | § 2.4 |
| Lemma M3 (Abel Summation Principle)                                                      | Proved here                    | § 4.3 |
| Theorem EXPLICIT: $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$                                | Proved here                    | § 4.4 |
| $W_1 = C_T \cdot \tilde{T}^+$ self-adjoint                                               | Proved here                    | § 3.4 |
| $W_1$ : no RH/HP; $C_T$ OLS-calibrated                                                   | Proved here                    | § 3.4 |
| $\mu_j \sim C_T/\gamma_{k(j)}$ , $r_1 = 0.950$                                           | <b>Numerical</b>               | § 3.1 |
| Eigenvector localization 0.45–0.99                                                       | <b>Numerical</b>               | § 3.2 |
| $\eta_{\text{orig}} > 0$ for $\kappa \leq 1009$                                          | <b>Numerical</b>               | § 5.1 |
| $\Delta_{\text{Burst}} \approx 3.11 > 0$ , $\kappa$ -invariant                           | <b>Numerical</b>               | § 5.2 |
| $W_1$ is NOT an asymptotic HP-operator                                                   | <b>Numerical</b>               | § 3.4 |
| Theorem 5.3(i)(ii) (equidist. vanishing)                                                 | <b>Conditional</b>             | § 5.3 |
| Theorem 5.3(iii): $\Delta(\kappa) \geq \Delta_{\text{Burst}} > 0$                        | <b>Numerical</b>               | § 5.3 |
| Theorem 5.3(iv): $\eta_{\text{orig}}(\kappa) \rightarrow \eta_{\infty} \approx 0.81 > 0$ | <b>Numerical</b>               | § 5.3 |
| $\eta_{\text{orig}} > 0$ analytically                                                    | <b>Open</b>                    | § 6   |
| Assumption A(a): $\zeta(1 + in\gamma_k) \neq 0$                                          | <b>Settled</b> (Hadamard 1896) | § 6.1 |
| Assumption A(b): equidistribution $\{\gamma_k \log p \bmod 2\pi\}$                       | <b>Open</b>                    | § 6.1 |

## 1.5 Relation to Existing Work

**Hilbert–Pólya setting.** A Hilbert–Pólya operator  $\mathcal{H}$  would satisfy  $\mathcal{H}\psi_k = \gamma_k\psi_k$  — the zero ordinates appear directly as eigenvalues. The Tehrani operator  $\tilde{T}$  has eigenvalues  $\mu_j \sim C_T/\gamma_{k(j)}$  (numerically); the zero ordinates appear as *localization centers* of eigenvectors, not as eigenvalues. This is a weaker but constructive relationship. The inverse  $W_1 = C_T \cdot \tilde{T}^+$  is a natural candidate for an HP-operator, but it fails (§ 2.4): the correlation  $r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$  falls to 0.16. We use the standard formulation of the Hilbert–Pólya setting — see e.g. the survey by Bombieri [1] — without adopting it as a postulate.

*Standard formulation (used throughout this paper):* “ $\tilde{T}$  is a resonance operator: the zero ordinates  $\gamma_k$  appear as localization centers of its eigenvectors, not as eigenvalues.

This distinguishes  $\tilde{T}$  from the classical Hilbert–Pólya setting.”

**Connes’ approach** [3]. The Connes spectral triple framework provides structural parallels; no formal connection between  $\tilde{T}$  and the Connes trace formula is established in this paper.

**Goldston–Gönek** [4]. Their exponential sum formula provides an independent confirmation of  $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$  proved in §4.4 (see Remark 4.7).

**Paper 3** [10]. The present paper is the direct sequel; §2.1 recollects all necessary objects.

## 1.6 Outline

Section 2 defines  $\tilde{T}$ , proves the algebraic spectral identity, and establishes that  $\tilde{T}$  is not a Hilbert–Pólya operator. Section 3 presents the spectral analysis of  $\tilde{T}$ : eigenvalue structure, eigenvector localization, and the two arithmetic constants  $C_\eta$  and  $C_T$ . Section 4 develops the monotonicity mechanism: Abel summation (Lemma M3) and the prime exponential sum bound (Theorem EXPLICIT). Section 5 collects results on  $\eta_{\text{orig}} > 0$ : numerical confirmation, the  $\Delta$ -decomposition, and the conditional theorem. Section 6 states the open problems. Section 7 presents the Fisher-information geometric interpretation (all statements in §7 are explicitly marked as geometric model or heuristic).

## 2 The Tehrani Operator

### 2.1 Recollection: $H_{\text{str}}$ , $H_{\text{null}}$ , $\Phi$ , $T$

We fix throughout:

- A prime cutoff  $\kappa > 1$ , with active prime set  $S(\kappa) := \{p \text{ prime} : p \leq \kappa\}$  and  $\pi(\kappa) := |S(\kappa)|$ .
- The first  $N$  positive imaginary parts of nontrivial zeros of  $\zeta(s)$ , ordered  $0 < \gamma_1 \leq \gamma_2 \leq \dots \leq \gamma_N$ .
- A Gaussian damping parameter  $\varepsilon > 0$ .

**Normative parameters:**  $\kappa = 53$ ,  $\varepsilon = 0.05$ ,  $N = 100$ ,  $\sigma = \frac{1}{2}$ .

**Definition 2.1** ( $H_{\text{str}}$  and  $H_{\text{null}}$ , from Paper 3).

$$H_{\text{str}} := \ell^2(S(\kappa)), \quad \dim H_{\text{str}} = \pi(\kappa). \quad H_{\text{null}} := \ell^2(\{\gamma_k\}_{k=1}^N), \quad \dim H_{\text{null}} = N.$$

*Both spaces carry standard inner products. We work throughout with real vectors and real operators.*

**Definition 2.2** (Resonance vectors, from Paper 3). *For each prime  $p \in S(\kappa)$ , the resonance vector  $a_p \in H_{\text{null}}$  is defined by*

$$a_p := \left( e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p) \right)_{k=1}^N.$$

The Gaussian envelope ensures  $a_p \in \ell^2(\{\gamma_k\})$  uniformly in  $\kappa$ .

**Definition 2.3** (Linear map  $\Phi$ , from Paper 3).

$$\Phi: H_{\text{str}} \rightarrow H_{\text{null}}, \quad \Phi(e_p) = a_p, \quad \Phi(c) = \sum_{p \leq \kappa} c_p a_p.$$

**Definition 2.4** (Loop operator  $T$ , from Paper 3).

$$T := \Phi^* \circ \Phi: H_{\text{str}} \rightarrow H_{\text{str}}, \quad T_{pq} = \langle a_p, a_q \rangle_{H_{\text{null}}} = G_{pq}^{\text{un}}.$$

$T$  is self-adjoint, positive semi-definite, and identical to the Gram matrix  $G^{\text{un}}$  of the resonance vectors.

**Definition 2.5** (Canonical weight vector, from Paper 3). The normative weight vector  $c \in H_{\text{str}}$  has components  $c_p := \sqrt{V_p(\frac{1}{2})}$ , where  $V_p(\sigma) = \frac{d^2}{d\sigma^2} \log \|M_\sigma f_{\kappa,p}\|^2$ . The exact formula is  $c_p = \sqrt{4(\log p)^2(2p-1)/(p(p-1)^2)}$ . The leading-term ( $K=1$ ) approximation  $c_p \approx 4\log(p)/\sqrt{p}$  is asymptotically close but numerically differs for small primes; the exact formula is used throughout this paper and in all verification scripts.

Reference values ( $\varepsilon = 0.05$ ,  $\sigma = \frac{1}{2}$ ):  $c_2 \approx 1.698$ ,  $c_3 \approx 1.418$ ,  $c_5 \approx 1.080$ ,  $c_7 \approx 0.884$ .

**Definition 2.6** (Energy quantities, from Paper 3).

$$E_{\text{str}} := \sum_{p \leq \kappa} c_p^2 \|a_p\|^2 = c^T D c, \quad E_{\text{spec}} := \left\| \sum_{p \leq \kappa} c_p a_p \right\|^2 = \|\Phi c\|^2,$$

where  $D := \text{diag}(\|a_p\|^2)_{p \in S(\kappa)}$ . The energy asymmetry is

$$\Delta := E_{\text{str}} - E_{\text{spec}} \geq 0 \quad (\text{numerically}), \quad \eta_{\text{orig}} := \frac{\Delta}{E_{\text{str}}} = 1 - \frac{c^T G^{\text{un}} c}{E_{\text{str}}}.$$

## 2.2 Definition of the Tehrani Operator $\tilde{T}$

**Definition 2.7** (Tehrani Operator).

$$\boxed{\tilde{T} := \Phi \circ \Phi^*: H_{\text{null}} \rightarrow H_{\text{null}}.} \tag{1}$$

**Matrix representation.** In the standard basis  $\{f_k\}_{k=1}^N$  of  $H_{\text{null}}$ :

$$\tilde{T}_{kl} = \sum_{p \leq \kappa} (a_p)_k (a_p)_l = \sum_{p \leq \kappa} e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p) \cdot e^{-\varepsilon^2 \gamma_l^2 / 2} \sin(\gamma_l \log p).$$

**Proposition 2.8** (Properties of  $\tilde{T}$ ). The operator  $\tilde{T}$  satisfies:

(i) **Self-adjointness:**  $\tilde{T}^* = (\Phi \Phi^*)^* = \Phi \Phi^* = \tilde{T}$ . ✓

(ii) **Positive semi-definiteness:** For all  $u \in H_{\text{null}}$ ,  $\langle \tilde{T}u, u \rangle = \|\Phi^* u\|^2 \geq 0$ . ✓

(iii) **Rank:**  $\text{rank}(\tilde{T}) = \text{rank}(\Phi) = \pi(\kappa)$ . ✓

(iv) **Explicit construction:**  $\tilde{T}$  is built directly from prime resonance vectors  $\{a_p\}$ . No zero ordinates appear as inputs to the operator itself; the  $\gamma_k$  enter only as axis labels of  $H_{\text{null}}$ .

Proof. Parts (i) and (ii) are immediate from  $\tilde{T} = \Phi\Phi^*$  using standard properties of adjoints. Part (iii) follows from  $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$  and  $\pi(\kappa) \leq N$  in all normative parameter choices. □

## 2.3 Algebraic Spectral Identity

**Theorem 2.9** (Spectral Identity PROVED).

$$\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}.$$

The shared nonzero eigenvalues satisfy:

$$\max_{j=1}^{\pi(\kappa)} |\lambda_j(T) - \mu_j(\tilde{T})| < 10^{-15} \quad (\text{machine precision, verified numerically}).$$

Proof. Classical result: for any  $A, B$ ,  $\sigma(AB) \setminus \{0\} = \sigma(BA) \setminus \{0\}$ . Applied with  $A = \Phi^*$ ,  $B = \Phi$ :  $T = \Phi^*\Phi$  and  $\tilde{T} = \Phi\Phi^*$ , so  $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$ . (For finite-dimensional matrices, Sylvester's determinant identity:  $\det(\lambda I - AB) = \lambda^{n-m} \det(\lambda I - BA)$  for  $A \in \mathbb{R}^{m \times n}$ ,  $B \in \mathbb{R}^{n \times m}$ .) □

**Remark.** The identity  $\lambda_j(T) = \mu_j(\tilde{T})$  is an algebraic theorem — it holds in exact arithmetic for any choice of  $\kappa$ ,  $\varepsilon$ , and  $N$ . The eigenvalues  $\lambda_j = \mu_j$  are not the zero ordinates  $\gamma_k$ ; they are constructive spectral quantities arising from  $\Phi$ .

## 2.4 $\tilde{T}$ is NOT a Hilbert–Pólya Operator

This section records a negative result.

**Proposition 2.10** (Numerical Negative Result). A Hilbert–Pólya operator  $\mathcal{H}$  would satisfy  $\mathcal{H}\psi_k = \gamma_k\psi_k$  for all  $k$ , i.e., the zero ordinates  $\gamma_k$  appear directly as eigenvalues.  $\tilde{T}$  does **not** have this property:

- (a) [Numerical] The eigenvalues of  $\tilde{T}$  satisfy  $\mu_j \sim C_T/\gamma_{k(j)}$  — they are inversely proportional to zero ordinates (§3.1).
- (b) [Numerical] The zero ordinates  $\gamma_k$  appear as localization centers of eigenvectors  $v_j$  (§3.2), not as eigenvalues.
- (c) [Numerical] The inverse operator  $W_1 := C_T \cdot \tilde{T}^+$  has eigenvalues  $\omega_j = C_T/\mu_j \approx \gamma_{k(j)}$  for small  $j$ , but the correlation  $r_2 := \text{corr}(\omega_j, \gamma_{k(j)})$  falls from 0.502 (at  $\kappa = 23$ ) to 0.160 (at  $\kappa = 503$ ).  $W_1$  is therefore **not** an asymptotic HP-operator.

The fundamental distinction:

$$\text{HP-operator: eigenvalues} = \gamma_k. \quad \tilde{T}: \text{eigenvector localization centers} = \gamma_k.$$

$\tilde{T}$  is a **resonance operator**: it encodes the  $\gamma_k$  as mode centers, not as spectral values.

Proof of (a)(b)(c). Numerical; see § 3.1, § 3.2, § 3.4 for detailed data.  $\square$

**Remark** (Structural obstruction). The Gaussian damping  $e^{-\varepsilon^2 \gamma_k^2/2}$  in  $a_p$  causes  $\mu_j$  to decay super-exponentially (exponent  $\alpha \approx -4.88$ ), far faster than the  $1/\gamma$ -law (exponent  $-0.56$ ). Inverting  $\tilde{T}$  to form  $W_1$  catastrophically amplifies the smallest eigenvalues. For  $j = 16$  at  $\kappa = 53$ :  $\omega_{16} \approx 7 \times 10^6$  against  $\gamma_{16} \approx 67$ . This is the structural reason  $W_1$  fails as an HP-operator (see § 3.4 and Open Problem 6.3).

### 3 Spectral Analysis of $\tilde{T}$

#### 3.1 Eigenvalue Structure: $\mu_j \sim C_T/\gamma_{k(j)}$

**Numerical Result NR 3.1** (Eigenvalue–Zero Ordinate Correlation). [Status: NUMERICAL —  $\kappa = 53$ ,  $\varepsilon = 0.05$ ,  $N = 100$ ]

All numerical results in this paper are reproducible using the verification scripts at <https://github.com/utehrani/analysislab-nt> (public repository, March 2026) [2]. All computations use  $N = 100$  zero ordinates computed via `mpmath` at default precision (approximately 15 significant digits) and normative parameters  $\kappa = 53$ ,  $\varepsilon = 0.05$ ,  $\sigma = \frac{1}{2}$  unless otherwise stated.

Let  $k_{\text{dom}}(j) := \arg \max_k |(v_j)_k|$  be the dominant zero index for eigenvector  $v_j$ . Then:

| $j$ | $\mu_j(\tilde{T})$ | $k_{\text{dom}}$ | $\gamma_{k_{\text{dom}}}$ | Localization |
|-----|--------------------|------------------|---------------------------|--------------|
| 1   | 4.219803           | 1                | 14.1347                   | 0.985        |
| 2   | 2.565694           | 3                | 25.0109                   | 0.741        |
| 3   | 1.744649           | 2                | 21.0220                   | 0.745        |
| 4   | 0.986492           | 4                | 30.4249                   | 0.854        |
| 5   | 0.314199           | 5                | 32.9351                   | 0.794        |
| 6   | 0.217900           | 6                | 37.5862                   | 0.885        |

The OLS correlation between  $\mu_j$  and  $1/\gamma_{k(j)}$  is

$$r_1 = \text{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950 \quad (\kappa = 53).$$

For  $\kappa = 503$ :  $r_1 = 0.859$  (weakening as  $\kappa$  grows, due to Gaussian damping).

**Remark.** The dominant index satisfies  $k_{\text{dom}}(1) = 1$  for all tested  $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ : the first eigenvector of  $\tilde{T}$  is always associated with  $\gamma_1 = 14.1347$ .



**Remark** (Cauchy–Schwarz warning). *Note that  $r_1 = 0.950$  coexists with  $r_2 = 0.502$  (§ 3.4). These measure different things:  $r_1$  tests monotone ordering;  $r_2$  tests value-level approximation. The distinction between ordering and value accuracy is crucial for the HP assessment in § 2.4.*

### 3.2 Eigenvector Localization

**Numerical Result NR 3.2** (Eigenvector Localization). [*Status: NUMERICAL* —  $\kappa = 53$ ,  $\varepsilon = 0.05$ ,  $N = 100$ ]

For each eigenvector  $v_j$  of  $\tilde{T}$  (normalized:  $\|v_j\| = 1$ ), the localization is measured by  $L_j := |(v_j)_{k_{\text{dom}}(j)}|$ . Observed range:  $L_j \in [0.45, 0.99]$  for  $j = 1, \dots, 14$ .

For comparison, a uniformly random unit vector in  $\mathbb{R}^{100}$  has typical entry magnitude  $1/\sqrt{100} = 0.10$ . The observed localization is 4.5–9.9 times the random baseline.

[*Geometric interpretation, not used in any proof*]: The localization structure suggests that  $\tilde{T}$  ‘encodes’ the ordering of zero ordinates in a distributional sense: each mode  $v_j$  is concentrated near the zero  $\gamma_{k(j)}$  in the zero-space  $H_{\text{null}}$ . This is the precise content of calling  $\tilde{T}$  a resonance operator.

### 3.3 The Two Arithmetic Constants $C_\eta$ and $C_T$

**Definition 3.1** (Arithmetic Constants).

$$C_\eta := \lim_{\kappa \rightarrow \infty} \frac{\lambda_{\max}(T_{\text{ren}})}{\pi(\kappa)}, \quad C_\eta \approx 0.39,$$

where  $T_{\text{ren}} := D^{-1/2}TD^{-1/2}$ .

$$C_T(\kappa) := \text{OLS-slope of } \mu_j \sim C_T/\gamma_{k(j)}, \quad C_T/\pi(\kappa) \in \{2.49, 2.21, 2.17, 1.94, 1.89\}$$

for  $\kappa \in \{23, 53, 101, 199, 503\}$  (weakly decreasing).

$$\boxed{C_\eta \neq C_T}$$

These are two distinct arithmetic quantities. Always use subscripts.

The constant  $C_\eta \approx 0.39$  is  $\kappa$ -independent;  $C_T$  grows with  $\kappa$  (numerically observed to grow approximately as  $2\pi(\kappa)$ ).

**Numerical Result NR 3.3** ( $\lambda_{\max, \text{ren}}$  growth). [*Status: NUMERICAL*]

| $\kappa$ | $\pi(\kappa)$ | $\lambda_{\max, \text{ren}}$ | Ratio |
|----------|---------------|------------------------------|-------|
| 23       | 9             | 3.59                         | 0.399 |
| 53       | 16            | 6.38                         | 0.399 |
| 101      | 26            | 10.35                        | 0.398 |
| 199      | 46            | 17.76                        | 0.386 |
| 503      | 96            | 37.38                        | 0.389 |
| 1009     | 169           | 65.16                        | 0.386 |

Variance of the ratio:  $\sigma^2 < 0.0003$  — extremely stable. The universal bound  $\lambda_{\max, \text{ren}} < 1$  (an earlier conjecture in this series, now falsified) is numerically falsified:  $\lambda_{\max, \text{ren}} \rightarrow \infty$ .

**Remark** (Analytical status of  $C_\eta$ ). *The value  $C_\eta \approx 0.39$  is not analytically explained. A candidate formula  $C_\eta = 1/I_N(\varepsilon)$  is **falsified**: it gives  $1/I_{100}(0.05) \approx 0.73 \neq 0.39$ . The correct representation involves a harmonic mean (analytically open; see Open Problem 6.4 and §6).*

### 3.4 The Inverse Operator $W_1 = C_T \cdot \tilde{T}^+$

**Definition 3.2** (Inverse Operator).  $W_1 := C_T \cdot \tilde{T}^+$ , where  $\tilde{T}^+$  denotes the Moore–Penrose pseudoinverse of  $\tilde{T}$  and  $C_T$  is the OLS constant from Definition 3.1.

**Proposition 3.3** (Self-Adjointness of  $W_1$  — PROVED).  $W_1$  is self-adjoint.

Proof.  $\tilde{T}$  is self-adjoint (Proposition 2.8(i)). For any self-adjoint operator, the Moore–Penrose pseudoinverse is also self-adjoint (standard result in linear algebra).  $C_T$  is a real scalar. Therefore  $W_1 = C_T \cdot \tilde{T}^+$  is self-adjoint.  $\square$

Numerically:  $\|W_1 - W_1^T\|_\infty = 3.1 \times 10^{-4}$  (machine precision effects from eigendecomposition).

**Proposition 3.4** (Non-Circularity Status of  $W_1$  — PROVED).  $W_1 = C_T \cdot \tilde{T}^+$  does not assume RH or the Hilbert–Pólya postulate. However, its normalization constant  $C_T$  is the OLS slope in  $\mu_j \sim C_T/\gamma_{k(j)}$ , which is calibrated against the zero ordinates.  $W_1$  should therefore not be presented as fully input-free.  $\tilde{T}$  itself — built from prime resonance vectors without  $\gamma_k$  as algebraic inputs — is the circularity-free object.

Proof.  $\tilde{T}$  is constructed from prime resonance vectors alone (Definition 2.7)  $\checkmark$ .  $\tilde{T}^+$  is the Moore–Penrose pseudoinverse — no postulate  $\checkmark$ . The scalar  $C_T$  is computed via OLS regression of  $\mu_j$  against  $1/\gamma_{k(j)}$ : this uses  $\gamma_k$  as calibration values, but does not assume RH or HP.  $\square$

**Numerical Result NR 3.4** ( $W_1$  HP-correlation — failure). [Status: NUMERICAL]

| $\kappa$ | $\pi(\kappa)$ | $r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$ |
|----------|---------------|----------------------------------------------|
| 23       | 9             | 0.502                                        |
| 53       | 16            | 0.436                                        |
| 101      | 26            | 0.321                                        |
| 199      | 46            | 0.226                                        |
| 503      | 96            | 0.160                                        |

The decreasing trend confirms:  $W_1$  is **not** an asymptotic HP-operator. The structural cause is Gaussian damping:  $e^{-\varepsilon^2 \gamma_k^2/2}$  in  $\Phi$  causes superexponential decay of  $\mu_j$  (observed exponent  $\alpha \approx -4.88$ ), far exceeding the  $1/\gamma$  growth pattern required for HP. Pseudoinversion catastrophically amplifies tail modes.

**Summary for § 3.4:**  $W_1$  is self-adjoint (proved). Its construction does not assume RH or the HP postulate, but  $C_T$  is calibrated via OLS on the zero ordinates —  $W_1$  is therefore not fully input-free (Proposition 3.4).  $W_1$  is the natural HP-candidate but fails numerically as  $\kappa$  grows. This rules out this particular pseudoinverse construction as an HP candidate. This does not preclude other spectral functions  $f(\tilde{T})$  from being valid HP-candidates; see Open Problem 6.3.

## 4 The Monotonicity Mechanism

The strategy for bounding  $\eta_{\text{orig}}$  from below rests on the interplay of three structural features: the monotone weight sequence  $(c_p)$ , the oscillatory mode structure  $(\sin(\gamma_k \log p))$ , and Abel summation.

### 4.1 Monotone Weight Structure (Lemma M1) — [Partial]

**Lemma 4.1** (Monotone Weight Structure). [Status: PROVED for  $p \geq 11$ ,  $K = 1$ ; general case OPEN]

For  $K = 1$  (leading term of  $V_p(\sigma)$ ) and  $p \geq 11$ :

$$c_p \approx \frac{4 \log p}{\sqrt{p}} \quad \text{is strictly monotone decreasing in } p.$$

Proof. Differentiate with respect to  $p$  (treating  $p$  as continuous):  $\frac{d}{dp}[\log p / \sqrt{p}] = (1/2 - \log p)/(p\sqrt{p}) < 0$  for  $p > e^{1/2} \approx 1.65$ . This holds for all primes  $p \geq 2$ .  $\square$

**Remark** (Status for  $K > 1$ ). When  $K > 1$  (i.e.,  $p^2 \leq \kappa$ ), the weight formula  $V_p(\sigma, K)$  is more complex. The monotonicity  $V_p(\sigma, K+1) \geq V_p(\sigma, K)$  is numerically supported but not formally proved. The general monotonicity of  $c_p$  for all  $p$  with  $K > 1$  is therefore OPEN.

### 4.2 Oscillatory Mode Structure (Lemma M2) — [Partial]

**Lemma 4.2** (Oscillatory Mode Structure). [Partial status — see below]

(M2a) [PROVED]: The function  $p \mapsto \sin(\gamma_k \log p)$  changes sign in the intervals

$$I_m := [e^{m\pi/\gamma_k}, e^{(m+1)\pi/\gamma_k}], \quad m = 1, 2, 3, \dots$$

Proof. Direct from  $\sin(x) = 0$  at  $x = m\pi$ : the argument  $\gamma_k \log p$  crosses  $m\pi$  at  $p = e^{m\pi/\gamma_k}$ .  $\square$

(M2b) [Partial]: By the Prime Number Theorem, each interval  $I_m$  contains at least one prime  $p \leq \kappa$ , for  $m$  in a range depending on  $\gamma_k$  and  $\kappa$ . Full quantitative control for all  $m$  is OPEN.

### 4.3 Abel Summation Principle (Lemma M3) — [PROVED]

**Lemma 4.3** (Abel Summation Principle — PROVED). *Let  $P \geq 1$ . Let  $(f_1, \dots, f_P)$  be monotone non-increasing and non-negative. Let  $(g_1, \dots, g_P)$  be any sequence with partial sums bounded by  $M$ :  $|G_m| := |\sum_{i=1}^m g_i| \leq M$  for all  $m$ . Then:*

$$\left| \sum_{i=1}^P f_i g_i \right| \leq M \cdot f_1.$$

Proof. Apply the Abel partial summation formula:  $\sum_{i=1}^P f_i g_i = f_P G_P - \sum_{i=1}^{P-1} (f_{i+1} - f_i) G_i$ . Taking absolute values:

$$\begin{aligned} \left| \sum_{i=1}^P f_i g_i \right| &\leq |f_P| \cdot |G_P| + \sum_{i=1}^{P-1} |f_{i+1} - f_i| \cdot |G_i| \\ &\leq M f_P + M \sum_{i=1}^{P-1} (f_i - f_{i+1}) \quad [f \text{ decreasing}] \\ &= M f_P + M(f_1 - f_P) = M f_1. \quad \square \end{aligned}$$

**Remark** (Application context for Paper 4). With  $f_i = (D^{-1/2} \tilde{c})_{p_i} \propto c_{p_i} / \|a_{p_i}\|$  (monotone decreasing — Lemma 4.1) and  $g_i = (u_j)_{p_i}$  (oscillatory eigenvector entries — Lemma 4.2), Lemma 4.3 bounds the projection  $|\langle u_j, D^{-1/2} \tilde{c} \rangle| \leq M_{k(j)}(\kappa) \cdot c_{p_1} / \|a_{p_1}\|$ , provided the partial sums of  $u_j$ -entries are bounded by  $M_{k(j)}(\kappa)$ . [The full reduction is structural, not a direct consequence of Lemma 4.3 alone.]

### 4.4 Prime Exponential Sum Bound (Theorem EXPLICIT) — [PROVED]

**Theorem 4.4** (Prime Exponential Sum Bound — PROVED — no RH used). *For fixed  $\gamma > 0$  and  $\kappa \geq 2$ :*

$$M_k(\kappa) := \max_{\substack{x \leq \kappa \\ p \leq x}} \left| \sum \sin(\gamma_k \log p) \right| = O\left( \frac{\pi(\kappa)}{\gamma_k} \right).$$

Explicitly:

$$M_k(\kappa) \leq \left( \frac{2}{\gamma_k} + o\left( \frac{1}{\gamma_k} \right) \right) \pi(\kappa), \quad \kappa \geq 2.$$

Proof. We establish three auxiliary lemmas.

**Lemma T1** (Abel–Stieltjes Representation — PROVED). *For  $\gamma > 0$  and  $\kappa \geq 2$ :*

$$\sum_{p \leq \kappa} p^{i\gamma} = A_0(\gamma) + J_\gamma(\kappa) + R_\gamma(\kappa),$$

where  $|A_0(\gamma)| \leq 0.1$ ;  $J_\gamma(\kappa) := \int_2^\kappa x^{i\gamma} / \log x \, dx$  (main term);  $R_\gamma(\kappa) := \kappa^{i\gamma} E(\kappa) - i\gamma \int_2^\kappa E(x) x^{i\gamma-1} dx$ , with  $E(x) = O(x e^{-c\sqrt{\log x}})$  (error term). No RH is used; PNT provides the  $E(x)$  bound.  $\square$

**Lemma T2** (Main Term Bound — PROVED).

$$|J_\gamma(\kappa)| \leq \left( \frac{2}{\gamma} + o\left(\frac{1}{\gamma}\right) \right) \pi(\kappa).$$

Proof. The dominant term satisfies  $|\kappa^{1+i\gamma}/((1+i\gamma)\log \kappa)| = \kappa/(\sqrt{1+\gamma^2} \cdot \log \kappa) \leq \kappa/(\gamma \log \kappa)$ . Since  $\pi(\kappa) \sim \kappa/\log \kappa$  (PNT):  $|J_\gamma(\kappa)| \leq (2/\gamma)\pi(\kappa)(1 + O(1/\log \kappa))$ .  $\square$

**Lemma T3** (Error Term Control — PROVED). For fixed  $\gamma > 0$  and  $\kappa \rightarrow \infty$ :  $|R_\gamma(\kappa)| = o(\pi(\kappa))$ .

Proof. PNT (Hadamard/de la Vallée-Poussin):  $|E(x)| = O(xe^{-c\sqrt{\log x}})$ . Substitution  $u = \sqrt{\log x}$  yields  $\int_2^\kappa e^{-c\sqrt{\log x}} dx = O(\sqrt{\log \kappa} \cdot \kappa \cdot e^{-c\sqrt{\log \kappa}}) = o(\pi(\kappa))$ , since  $\sqrt{\log \kappa} \cdot e^{-c\sqrt{\log \kappa}} = o(1/\log \kappa)$ . Combined:  $|R_\gamma(\kappa)| = o(\pi(\kappa))$  for fixed  $\gamma$ .  $\square$

Proof of Theorem 4.4: The imaginary part of Lemma T1 gives  $A_k(\kappa) = \text{Im}[A_0] + \text{Im}[J_{\gamma_k}] + \text{Im}[R_{\gamma_k}]$ . Applying T2 and T3:

$$|A_k(\kappa)| \leq 0.1 + \frac{2}{\gamma_k} \pi(\kappa)(1 + o(1)) + o(\pi(\kappa)) = O\left(\frac{\pi(\kappa)}{\gamma_k}\right).$$

The same bound holds for all partial sums (Remark 4.5), so the maximum  $M_k(\kappa)$  satisfies the same bound.  $\square$

**Corollary 4.5.** For  $k \rightarrow \infty$  (with  $\gamma_k \rightarrow \infty$ ):  $M_k(\kappa)/\pi(\kappa) \rightarrow 0$  uniformly in  $\kappa$ .

**Remark** (Goldston–Gönek confirmation [4]). An independent formula of Goldston–Gönek yields the same  $O(\pi(\kappa)/\gamma_k)$  bound via mean-value methods. This provides an independent check. It provides no improvement for fixed  $k$ , confirming that the bottleneck is the 1-dimensional equidistribution question Assumption A(a).

**Remark** (What Theorem EXPLICIT does NOT prove). Theorem 4.4 does not prove  $M_k(\kappa) = o(\pi(\kappa))$  for fixed  $k$  as  $\kappa \rightarrow \infty$ . This stronger statement is equivalent (via Weyl’s equidistribution criterion) to  $\zeta(1 + in\gamma_k) \neq 0$  for all  $n \geq 1$  — the 1-dimensional component Assumption A(a) of the joint assumption Assumption A (§6, Open Problem 6.1).

## 5 Results on $\eta_{\text{orig}} > 0$

### 5.1 Numerical Confirmation

**Numerical Result NR 5.1** ( $\eta_{\text{orig}} > 0$  for  $\kappa \leq 1009$ ). [Status: NUMERICAL —  $\varepsilon = 0.05$ ,  $N = 100$ ]

Let  $\tilde{c} = D^{1/2}c/\sqrt{E_{\text{str}}}$  be the normalized canonical weight vector. The Rayleigh quotient identity

$$\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle = 1 - \eta_{\text{orig}}$$

holds to machine precision ( $< 2 \times 10^{-16}$ ), where  $T_{\text{ren}} := D^{-1/2}TD^{-1/2}$ .

| $\kappa$ | $\pi(\kappa)$ | $\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle$ | $< 1?$ | $\eta_{\text{orig}}$ |
|----------|---------------|-------------------------------------------------------|--------|----------------------|
| 23       | 9             | 0.18849                                               | ✓      | 0.81151              |
| 53       | 16            | 0.33073                                               | ✓      | 0.66927              |
| 101      | 26            | 0.22803                                               | ✓      | 0.77197              |
| 199      | 46            | 0.21740                                               | ✓      | 0.78260              |
| 503      | 96            | 0.21788                                               | ✓      | 0.78212              |
| 1009     | 169           | 0.18002                                               | ✓      | 0.81998              |

$\eta_{\text{orig}} > 0$  for all tested  $\kappa \leq 1009$  ✓.

**Collective mechanism.** The dominant contribution to  $\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle = \sum_j B_j$  comes from mode  $j = 4$  ( $B_4 = 0.193$ ), not from the dominant eigenmode  $j = 1$  ( $B_1 = 0.040$ ). Approximately 92% cancellation occurs at  $j = 1$  ( $\kappa = 53$ ). The effect is distributed across  $\sim 8$  eigenmodes —  $\eta_{\text{orig}} > 0$  is a **collective spectral effect**, not a simple  $u_1$ -orthogonality principle.

## 5.2 $\Delta$ -Decomposition and Burst Stability

**Definition 5.1** ( $\Delta$ -Decomposition). *With  $S(r) := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log r)$ :*

$$\Delta = \Delta_{\text{Burst}} + \Delta_{\text{Cross}} + \Delta_{\text{Stream}},$$

where

$$\Delta_{\text{Burst}} = \sum_{p < q \leq 7} c_p c_q [S(pq) - S(p/q)], \quad \Delta_{\text{Cross}} = \sum_{p \leq 7 < q \leq \kappa} c_p c_q [S(pq) - S(p/q)],$$

$$\Delta_{\text{Stream}} = \sum_{7 < p < q \leq \kappa} c_p c_q [S(pq) - S(p/q)].$$

The factorization  $S(pq) - S(p/q) = -2 \sum_k e^{-\varepsilon^2 \gamma_k^2} \sin(\gamma_k \log p) \sin(\gamma_k \log q)$  follows from  $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$ . [PROVED, algebraic]

**Numerical Result NR 5.4** ( $\Delta_{\text{Burst}}$  stability). [NUMERICAL]

| Pair         | $S(pq)$ | $S(p/q)$ | $c_p c_q$       | Contribution             |
|--------------|---------|----------|-----------------|--------------------------|
| (2, 3)       | 0.9536  | 0.2653   | $\approx 2.408$ | +1.657                   |
| (2, 5)       | 0.4110  | 0.6555   | $\approx 1.833$ | −0.448                   |
| (2, 7)       | 0.5794  | 0.5812   | $\approx 1.501$ | −0.003                   |
| (3, 5)       | 0.9440  | 0.3686   | $\approx 1.531$ | +0.881                   |
| (3, 7)       | 0.7193  | 0.5571   | $\approx 1.254$ | +0.203                   |
| (5, 7)       | 0.9668  | 0.1135   | $\approx 0.954$ | +0.814                   |
| <b>Total</b> |         |          |                 | $\approx +\mathbf{3.11}$ |

$\Delta_{\text{Burst}} \approx 3.11 > 0$ , and it is  $\kappa$ -**invariant** for all tested  $\kappa > 7$ . The sufficient condition for  $\Delta > 0$

is:

$$|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}| < \Delta_{\text{Burst}} \approx 3.11.$$

This is numerically satisfied for all  $\kappa \leq 1009$ ; analytic proof is OPEN.

### 5.3 Conditional Theorem (Theorem 5.3)

**Theorem 5.2** (Conditional and Numerical Results for  $\eta_{\text{orig}} > 0$ ). **Assumption A:**

(a) [**SETTLED** — *Hadamard-de la Vallée Poussin 1896*]:  $\zeta(1 + in\gamma_k) \neq 0$  for all integers  $n \geq 1$  and all  $k \leq N$ . This follows directly from Hadamard's theorem, which establishes  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$ ; the values  $t = n\gamma_k$  are real and positive, hence covered unconditionally.

(b) The sequences  $\{(\gamma_{k_1} \log p, \gamma_{k_2} \log p) \bmod 2\pi\}_{p \leq \kappa, \kappa \rightarrow \infty}$  are jointly equidistributed in  $[0, 2\pi)^2$  for all  $k_1 \neq k_2$  with  $k_1, k_2 \leq N$ .

Structural motivation: Part (a) is precisely what Weyl's equidistribution criterion requires to make the diagonal sums  $\frac{1}{\pi(\kappa)} \sum_p e^{in\gamma_k \log p}$  vanish — equivalently, it is the non-vanishing condition for  $\zeta$  on  $\text{Re}(s) = 1$  at the specific values  $s = 1 + in\gamma_k$ . Part (b) is the natural 2-dimensional extension required to make the off-diagonal (cross) terms  $\frac{1}{\pi(\kappa)} \sum_p \sin(\gamma_{k_1} \log p) \sin(\gamma_{k_2} \log p)$  vanish for  $k_1 \neq k_2$ ; without it, the cross-term cancellation in step (ii) cannot be established. These two conditions are the minimal structural requirements for the proof of (i)–(ii).

Note: Part (a) is settled by Hadamard (1896):  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$  covers  $t = n\gamma_k$  directly. Only Part (b) is genuinely open.

**Remark** (Relation between Part (a) and Weyl's criterion). Part (a) of Assumption A is settled:  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$  (Hadamard-de la Vallée Poussin, 1896), covering  $t = n\gamma_k$ . Part (b) — the equidistribution of  $\{\gamma_k \log p \bmod 2\pi\}$  via Weyl's criterion — requires the vanishing of

$$\frac{1}{\pi(\kappa)} \sum_{p \leq \kappa} p^{in\gamma_k} \rightarrow 0 \quad \text{as } \kappa \rightarrow \infty, \quad \text{for all } n \geq 1, k \leq N.$$

This depends on the quantitative behaviour of  $\zeta$  near  $s = 1 + in\gamma_k$ : while non-vanishing (Part (a)) is known, the quantitative rate sufficient for Weyl's criterion remains open. The genuine open content of Assumption A is therefore Part (b) alone.

Under this Assumption:

(i) [CONDITIONAL on Assumption A(b)] The sequence  $\{\gamma_k \log p \bmod 2\pi\}_{p \leq \kappa, \kappa \rightarrow \infty}$  is equidistributed modulo  $2\pi$  for each fixed  $k \leq N$ .

(ii) [CONDITIONAL on Assumption A(b) — Unweighted] The normalized unweighted cross averages vanish:

$$\frac{1}{\pi(\kappa)} \sum_{p \leq \kappa} \sin(\gamma_{k_1} \log p) \sin(\gamma_{k_2} \log p) \rightarrow 0 \quad \text{for all } k_1 \neq k_2.$$

(iii) [Numerical lower bound — no assumption needed]  $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$  for all tested  $\kappa > 7$ . [NUMERICAL —  $\kappa$ -invariant by construction.] Numerically, both  $\Delta_{\text{Cross}}$  and  $\Delta_{\text{Stream}}$  appear to converge individually, and their sum tends to  $\approx 1.8$ ; hence  $\Delta(\kappa) = \Delta_{\text{Burst}} + \Delta_{\text{Cross}} + \Delta_{\text{Stream}} \rightarrow \Delta_{\infty} \approx 3.11 + 1.8 \approx 4.9 > \Delta_{\text{Burst}}$ . The equidistribution of parts (i)–(ii) implies  $(\Delta_{\text{Cross}} + \Delta_{\text{Stream}})/\pi(\kappa) \rightarrow 0$ , but the absolute values converge to a finite positive limit, not to zero.

(iv)  $\eta_{\text{orig}}(\kappa) \rightarrow \eta_{\infty} = \Delta_{\infty}/E_{\text{str}}(\infty) > 0$ . [NUMERICAL:  $\eta_{\infty} \approx 0.81$ , consistent with  $\Delta_{\infty} \approx 4.9$  and  $E_{\text{str}}(\infty) \approx 6.1$ .] The lower bound from (iii) gives  $\eta_{\infty} \geq \Delta_{\text{Burst}}/E_{\text{str}}(\infty) \approx 0.51 > 0$  unconditionally [NUMERICAL].

Proof sketch. (i) follows from Assumption (a) by Weyl’s criterion (cf. [5], §21). (ii) follows from Assumption (b) via  $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$ . (iii) is a direct numerical observation:  $\Delta_{\text{Burst}}$  is  $\kappa$ -invariant by construction, and  $\Delta(\kappa) \geq \Delta_{\text{Burst}} > 0$  for all tested  $\kappa > 7$ . (iv) follows from  $\Delta_{\infty} > 0$  (numerically) and  $E_{\text{str}}(\infty) \approx 6.1 < \infty$  (proved). No RH is used.  $\square$

**Remark** (Weighted transfer — correction note). The equidistribution assumption Assumption A implies  $\Delta_{\text{Stream}}(\kappa)/\pi(\kappa) \rightarrow 0$  and  $\Delta_{\text{Cross}}(\kappa)/\pi(\kappa) \rightarrow 0$  (Cross/Stream terms are  $o(\pi(\kappa))$ ). However, numerically  $\Delta_{\text{Cross}} + \Delta_{\text{Stream}}$  converges to a finite positive limit  $\approx 1.8$ , not to zero. The earlier formulation “ $\Delta(\kappa) \rightarrow \Delta_{\text{Burst}}$ ” was therefore incorrect:  $\Delta_{\text{Burst}} \approx 3.11$  is a  $\kappa$ -**invariant lower bound** for  $\Delta(\kappa)$ , not the limit. The correct limit is  $\Delta_{\infty} \approx 4.9 > \Delta_{\text{Burst}}$ . The conclusion  $\eta_{\infty} > 0$  follows from the numerical lower bound  $\Delta(\kappa) \geq \Delta_{\text{Burst}} > 0$ , without requiring the equidistribution assumption.

**Remark** (Status of the Assumption — corrected). Theorem 5.2 requires Assumption A. The assumption concerns the line  $\text{Re}(s) = 1$ , distinct from the Riemann Hypothesis which concerns  $\text{Re}(s) = \frac{1}{2}$ . The logical relationship between Assumption A and RH is not analysed in this paper.

**What is known.**  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$  (Hadamard and de la Vallée Poussin, 1896). This covers all values  $t = n\gamma_k$  directly —  $n\gamma_k$  is a positive real number for  $n \geq 1$ ,  $k \leq N$ , and Hadamard’s theorem is unconditional on the structure of  $t$ . Part (a) of Assumption A therefore **requires no proof**: it is a consequence of Hadamard’s theorem.

**What is open (OF-EXPLICIT-1’).** The equidistribution of  $\{\gamma_k \log p \bmod 2\pi\}$  on the circle  $[0, 2\pi)$ . Via Weyl’s criterion, this is equivalent to:

$$\frac{1}{\pi(\kappa)} \sum_{p \leq \kappa} e^{in\gamma_k \log p} \rightarrow 0 \quad \text{as } \kappa \rightarrow \infty, \quad \text{for all } n \geq 1, k \leq N.$$

This is a statement about the average behaviour of primes, not about individual values of  $\zeta$ . It is not implied by Hadamard’s theorem alone. The genuine open content of Assumption A is Part (b): equidistribution.

## 5.4 Reduction to $\text{Re}(s) = 1$

**Remark** (The line  $\text{Re}(s) = 1$  vs.  $\text{Re}(s) = \frac{1}{2}$ ). Theorem 5.2 requires Assumption A. Part (a) translates via Weyl’s criterion into a non-vanishing condition:  $\zeta(1 + in\gamma_k) \neq 0$  for all  $n \geq 1$ ,



$k \leq N$ . This concerns  $\operatorname{Re}(s) = 1$  — **not**  $\operatorname{Re}(s) = \frac{1}{2}$ .

Within the present program, the implication chain is:

$\text{RH} \longleftarrow [\text{transfer } \sigma = \frac{1}{2} \rightarrow \sigma \neq \frac{1}{2}, \text{ open}] \longleftarrow \eta_{\text{orig}} > 0 \longleftarrow \text{Theorem 5.2} \longleftarrow \text{Assumption A (open)}.$

Paper 4 proves the conditional implication in the middle. The connection to RH requires a quantitative transfer from  $\sigma = \frac{1}{2}$  to  $\sigma \neq \frac{1}{2}$  on the curvature level — a step entirely outside the scope of this paper.

## 6 Open Problems

**Open Problem** (Assumption A — Main Open Problem). The proof of Theorem 5.2 requires:

(a)  $\zeta(1 + in\gamma_k) \neq 0$  for all  $n \geq 1$ ,  $k \leq N$ .

Note: Part (a) alone — the non-vanishing  $\zeta(1 + in\gamma_k) \neq 0$  — follows directly from Hadamard's theorem (1896), which establishes  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$ . The values  $t = n\gamma_k$  are real and positive, hence covered. The genuine open content of Assumption A is Part (b): the equidistribution of  $\{\gamma_k \log p \bmod 2\pi\}$  via Weyl's criterion.

(b) Joint equidistribution of  $\{(\gamma_{k_1} \log p, \gamma_{k_2} \log p) \bmod 2\pi\}$  for all  $k_1 \neq k_2$ ,  $k_1, k_2 \leq N$ .

Known for (a):  $\zeta(1 + it) \neq 0$  for all real  $t \neq 0$  (Hadamard, 1896); this covers  $t = n\gamma_k$  directly. Condition (b) is independent and open.

Possible approaches: (A) Direct Hadamard product estimate near  $s = 1 + in\gamma_k$ . (B) Linear independence of  $\{\gamma_k \log p\}$  over  $\mathbb{Q}$  (sufficient for (b)). (C) Numerical verification for small  $n$ ,  $k$ .

**Open Problem** (Analytic proof of  $\eta_{\text{orig}} > 0$  without Assumption). Prove  $\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle < 1$  for the canonical weight system without the equidistribution assumption of Theorem 5.2.

Current best:  $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$  (Theorem 4.4) bounds individual projections but is insufficient for the full sum  $\sum_j \lambda_j |\langle u_j, D^{-1/2} \tilde{c} \rangle|^2 < E_{\text{str}}$ .

**Open Problem** (HP-operator — negative result, circularity-free). Does any circularity-free spectral function  $f(\tilde{T})$  satisfy  $\operatorname{Spec}(f(\tilde{T})) \rightarrow \{\gamma_k\}$  as  $\kappa \rightarrow \infty$ ?

The natural candidate  $W_1 = C_T \cdot \tilde{T}^+$  fails:  $r_2 \rightarrow 0.16$  (§3.4). The structural obstruction is Gaussian damping.

A possible modification: replace  $e^{-\varepsilon^2 \gamma_k^2/2}$  by  $e^{-\varepsilon \gamma_k}$  (slower decay) in  $\Phi$ .

**Open Problem** (Arithmetic constant  $C_\eta \approx 0.39$ ). Determine the analytic formula for  $C_\eta = \lim_{\kappa \rightarrow \infty} \lambda_{\text{max,ren}}/\pi(\kappa)$ . The candidate  $C_\eta = 1/I_N(\varepsilon)$  is falsified. The correct representation involves a harmonic mean (analytically open).

**Open Problem** (Asymptotics of  $C_T(\kappa)/\pi(\kappa)$ ). Does  $C_T(\kappa)/\pi(\kappa)$  converge as  $\kappa \rightarrow \infty$ ? Numerically: values 2.49, 2.21, 2.17, 1.94, 1.89 (decreasing) — the limit, if it exists, is analytically undetermined.

**Open Problem** (Monotone bound). *Is  $\eta_{\text{orig}} > 0$  a consequence of the Prime Number Theorem alone, independent of any RH-related hypothesis?*

*Evidence:*  $\Delta_{\text{Burst}} \approx 3.11 > 0$  depends only on primes  $p \leq 7$  and is  $\kappa$ -invariant. Analytically OPEN.

**Remark** (Connection to RH — Geometric Interpretation, not a theorem). [Geometric interpretation — not established as a formal statement in this paper.]

*The implication chain:*

$$\text{RH} \leftarrow [\text{transfer open}] \leftarrow \eta_{\text{orig}} > 0 \leftarrow \text{Theorem 5.2} \leftarrow \zeta(1 + in\gamma_k) \neq 0.$$

*No direct connection between  $\eta_{\text{orig}} > 0$  and RH is established in this paper. The missing step is a quantitative transfer from  $\sigma = \frac{1}{2}$  to  $\sigma \neq \frac{1}{2}$  on the curvature level — a step entirely outside the scope of this paper and the subject of future work.*

## 7 Fisher-Information Connection (Geometric Interpretation)

*All statements in this section are clearly marked as geometric interpretation or [Geometric heuristic — not used in proof]. None are used in any argument in §§ 2–5.*

### 7.1 The Weight Vector as Fisher Information

The canonical weight  $c_p = \sqrt{V_p(\frac{1}{2})}$  has the structure of a Fisher information weight at  $\sigma = \frac{1}{2}$ :

*[Geometric interpretation]:*  $V_p(\sigma) = \frac{d^2}{d\sigma^2} \log \|M_\sigma f_{\kappa,p}\|^2$  is the log-convexity of the local norm squared — analogous to Fisher information for a location family. The canonical weight  $c_p$  assigns larger weight to primes that are “more informative” about  $\sigma$  at  $\sigma = \frac{1}{2}$ .

### 7.2 Energy Interpretation

*[Geometric interpretation]:*

$$E_{\text{str}} = \sum_p V_p(\tfrac{1}{2}) \|a_p\|^2 \approx \text{“Fisher-weighted resonance energy”}.$$

$$\Delta = \text{Var}(\sqrt{V_p(\tfrac{1}{2})} \cdot a_p) \approx \text{“spectral variance of Fisher-weighted resonance”}.$$

*[Geometric heuristic — not a mathematical claim]:*  $\eta_{\text{orig}}$  may be interpreted as a “coding efficiency of primes for zero localization”.

### 7.3 Cramér–Rao Analogy

*[Geometric heuristic — not used in proof — not used in any proof]:*

The Cramér–Rao inequality reads  $\text{Var}(\hat{\theta}) \geq 1/\mathcal{I}(\theta)$ . By structural analogy:  $E_{\text{str}} \longleftrightarrow \mathcal{I}(\frac{1}{2})$ ;  $\Delta \longleftrightarrow \text{Var}$ .

[*Speculative analogy — not a theorem*]: A Cramér–Rao-type inequality  $\eta_{\text{orig}} \cdot H_{\text{local}} \geq C$  would be a quantitative information-theoretic bound; its existence is not established here. Numerically,  $\eta_{\text{orig}} \approx 0.67$  and  $H_{\text{local}} \sim 8(\log \kappa)^2 \rightarrow \infty$  would give a vanishing lower bound — consistent but non-informative. No formal version of this analogy has been proved.

*This section provides intuition for future work, not a proof strategy.*

## Acknowledgments

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*Pauca sed matura.* — C.F. Gauss

## Call for Feedback

This paper is part of the ongoing *AnalysisLab\_L1–L5* project, an open research initiative toward the Riemann Hypothesis. All papers are freely available on Zenodo under CC BY 4.0, and the accompanying code is hosted on GitHub.

The authors welcome critical feedback, corrections, and collaboration:

- **Zenodo:** <https://zenodo.org/search?q=Ulrich+Tehrani>
- **GitHub:** <https://github.com/utehrani/analysislab-nt>

In particular, we are interested in:

- Analytical approaches to the Weyl equidistribution of  $\{\gamma_k \log p \bmod 2\pi\}$  (Open Problem in §6)
- Connections to existing results on exponential sums over primes and zeros of  $\zeta(s)$
- Any errors or gaps in the present arguments

*Pauca sed matura.* (C.F. Gauss)

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## A Normative Parameter Table

| Parameter                         | Value                            | Definition                               |
|-----------------------------------|----------------------------------|------------------------------------------|
| $\kappa$                          | 53 (normative benchmark)         | Prime cutoff                             |
| $\varepsilon$                     | 0.05                             | Gaussian damping                         |
| $N$                               | 100                              | Number of zero ordinates                 |
| $\sigma$                          | 0.5                              | Reference point                          |
| $\eta_{\text{orig}}(\kappa = 53)$ | 0.66927                          | Energy asymmetry                         |
| $E_{\text{str}}(\infty)$          | $\approx 6.1$                    | Limiting structural energy               |
| $\eta_{\infty}$                   | $\approx 0.81 \pm 0.03$          | Limiting energy asymmetry                |
| $C_{\eta}$                        | $\approx 0.39$                   | $\lambda_{\text{max,ren}}/\pi(\kappa)$   |
| $C_T$ (normative)                 | $\approx 35.3$ ( $\kappa = 53$ ) | OLS slope $\mu_j \sim C_T/\gamma_{k(j)}$ |
| $\Delta_{\text{Burst}}$           | $\approx 3.11$                   | $\kappa$ -invariant for $\kappa > 7$     |
| $r_1$                             | 0.950 ( $\kappa = 53$ )          | $\text{corr}(\mu_j, 1/\gamma_{k(j)})$    |

## B Status Summary

| Result                                                                             | Section | Status                                   |
|------------------------------------------------------------------------------------|---------|------------------------------------------|
| $\tilde{T}$ self-adjoint, PSD                                                      | § 2.2   | PROVED                                   |
| Spectral identity $\lambda_j = \mu_j$                                              | § 2.3   | PROVED                                   |
| $\tilde{T} \neq$ HP-operator                                                       | § 2.4   | NUMERICAL NEGATIVE RESULT                |
| $W_1$ self-adjoint                                                                 | § 3.4   | PROVED                                   |
| $W_1$ : no RH/HP; $C_T$ OLS-calibrated                                             | § 3.4   | PROVED (Prop. 3.8)                       |
| Lemma M1 (monotone weights, $K = 1$ )                                              | § 4.1   | PROVED ( $p \geq 11$ ); OPEN ( $K > 1$ ) |
| Lemma M2a (sign changes)                                                           | § 4.2   | PROVED                                   |
| Lemma M3 (Abel summation)                                                          | § 4.3   | PROVED                                   |
| Theorem T1 (Abel–Stieltjes)                                                        | § 4.4   | PROVED                                   |
| Lemma T2 (main term bound)                                                         | § 4.4   | PROVED                                   |
| Lemma T3 (error term)                                                              | § 4.4   | PROVED                                   |
| Theorem EXPLICIT ( $M_k = O(\pi/\gamma_k)$ )                                       | § 4.4   | PROVED                                   |
| $\mu_j \sim C_T/\gamma_{k(j)}$ , $r_1 = 0.950$                                     | § 3.1   | NUMERICAL                                |
| Localization 0.45–0.99                                                             | § 3.2   | NUMERICAL                                |
| $W_1$ fails as HP-operator ( $r_2 \rightarrow 0.16$ )                              | § 3.4   | NUMERICAL                                |
| $\eta_{\text{orig}} > 0$ for $\kappa \leq 1009$                                    | § 5.1   | NUMERICAL                                |
| $\Delta_{\text{Burst}} \approx 3.11 > 0$                                           | § 5.2   | NUMERICAL                                |
| $C_\eta \approx 0.39$ structure                                                    | § 3.3   | NUMERICAL ( $1/I_N$ FALSIFIED)           |
| Thm 5.3(i)(ii): equidist. vanishing                                                | § 5.3   | CONDITIONAL (Assumption A(b))            |
| Thm 5.3(iii): $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$         | § 5.3   | NUMERICAL                                |
| Thm 5.3(iv): $\eta_{\text{orig}}(\kappa) \rightarrow \eta_\infty \approx 0.81 > 0$ | § 5.3   | NUMERICAL                                |
| Assumption A(a): $\zeta(1 + in\gamma_k) \neq 0$                                    | § 6     | SETTLED (Hadamard 1896)                  |
| Assumption A(b): equidistribution                                                  | § 6     | OPEN                                     |
| $\eta_{\text{orig}} > 0$ analytically                                              | § 6     | OPEN                                     |
| Closed HP-question for $W_1$                                                       | § 3.4   | CLOSED (falsified)                       |

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*v4: Priority-A fixes — Assumption A(a) correctly marked SETTLED (Hadamard); Weyl-bridge Remark added; Theorem 5.3 retitled CONDITIONAL/NUMERICAL; Call for Feedback added*

*Pauca sed matura*